

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 02



By- Vishal Sir



Recap of Previous Lecture

- ✓ **Topic** Sets and representation of sets
- ✓ **Topic** Types of sets
- ✓ **Topic** Number of subsets and supersets
- ✓ **Topic** Power set



Topics to be Covered



✓ Topic

Venn diagram

✓ Topic

Set operations and properties of set operations

✓ Topic

Multi-set



Topic : Power Set

Power Set: Let 'A' is any finite set, then power set
of set 'A' is a set of all subsets of set A.
Power set of set A is denoted by $P(A)$ or 2^A

$P(A)$ = Set of all subsets of set A

$P(A) = \{$
Set $\emptyset$ $\underbrace{\hspace{10em}}_{\text{Elements will be all subsets of set A}}$ $\}$

Q: Let $A = \{a, b, c\}$

then

$$P(A) = ?$$

Soln

$$P(A) = \{ \{ \}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$$

Set

Elements are
all subsets of Set A.

Note:-

① $P(A)$ is also a set, and we know that $\emptyset$ is a subset of Every set
 $\therefore \emptyset \subseteq P(A)$ is true for any set A .

② $\emptyset$ is always an element of $P(A)$, for any set A .

i.e., $\emptyset \in P(A)$ is true for any set A

③ For any set A
 $A \in P(A)$ is always true



Topic : Cardinality of Power Set

Cardinality
of $P(A)$ = $|P(A)|$ = No. of elements in $P(A)$
= Total no. of subsets of set A .
= $2^{|A|}$

$$|P(A)| = 2^{|A|}$$

Today's Topic

H.Wo Q: Let $\emptyset$ be an empty set, then

$$|P(P(\emptyset))| = ?$$

- (a) 0
- (b) 1
- ✓ (c) 2
- (d) Can not be defined.

$$|\emptyset| = 0$$

$$|P(A)| = 2^{|A|}$$

$$\downarrow$$
$$\therefore |P(\emptyset)| = 2^{|\emptyset|} = 2^0 = 1$$

$$\therefore |P(\underline{P(\emptyset)})| = 2^{|P(\emptyset)|} = 2^1 = 2$$

H.Wo Q: Let $\emptyset$ be an empty set, then

$$|P(P(\emptyset))| = ?$$

- (a) 0
- (b) 1
- ☒ (c) 2
- (d) Can not be defined.

$$\{\} \neq \{\{\}\}$$
$$\emptyset \neq \{\emptyset\}$$

$$P(\emptyset) = \{ \underbrace{\emptyset}_{\text{Subsets of } \emptyset} \}$$

$$P(P(\emptyset)) = \{ \emptyset, \underbrace{\{\emptyset\}}_{\text{Subsets of } P(\emptyset)} \}$$

2 elements

Let A is any set

$a \in A$ is true if and only if 'a' is an element of set A

$$\{a\} \subseteq A$$

Subset will be
a set containing
0 or more

elements from set A i.e. 'a' must be element of set A

HoW. Q: Let $A = \{ \underline{1}, \underline{2}, 3, \{1, 2\}, \{\{2, 3\}\} \}$

Which of the following is/are true

~~✓~~ (a) $1 \in A$

~~✗~~ (b) $\{1\} \in A$

~~✗~~ (c) $1 \subseteq A$

~~✓~~ (d) $\{1\} \subseteq A$

~~✓~~ (e) $\{1, 2\} \in A$

~~✓~~ (f) $\{1, 2\} \subseteq A$

~~✗~~ (g) $\{\{1, 2\}\} \in A$

~~✓~~ (h) $\{\{1, 2\}\} \subseteq A$

~~✗~~ (i) $\{2, 3\} \in A$

~~✓~~ (j) $\{2, 3\} \subseteq A$

~~✓~~ (k) $\{\{2, 3\}\} \in A$

~~✗~~ (l) $\{\{2, 3\}\} \subseteq A$

~~✓~~ (m) $\{\{\{2, 3\}\}\} \subseteq A$

~~✗~~ (n) $\emptyset \in A$

~~✓~~ (o) $\emptyset \in P(A)$

~~✓~~ (x) $\emptyset \subseteq A$

~~✓~~ (z) $\emptyset \subseteq P(A)$

subset must be a set

H.W.

$$P(A) = 2^A = \{ \emptyset, \{5\}, \{ \{6\} \}, \{ \{7\} \}, \{5, \{6\}\}, \{5, \{7\}\}, \{ \{6\}, \{7\} \}, \{5, \{6\}, \{7\}\} \}$$



Q2. For a set A , the power set of A is denoted by 2^A . If $A = \{5, \{6\}, \{7\}\}$ which of the following options are true?

✓ ~~1~~ $\emptyset \in 2^A$

✓ ~~2~~ $\emptyset \subseteq 2^A$

✓ ~~3~~ $\{5, \{6\}\} \in 2^A$

✗ ~~4~~ $\{5, \{6\}\} \subseteq 2^A$

How "PYQ"



Q1. Let $P(S)$ denote the power set of a set S . Which of the following is always true?

~~A.~~ $P(P(S)) = P(S)$ for any set S , $|P(S)| \neq |P(P(S))|$ $\therefore$ they ^{can} never be equal
(Always false)

~~B.~~ $P(S) \cap P(P(S)) = \{\emptyset\}$ $\leftarrow$ May be true or may be false
 $\{\emptyset, \dots\} \cap \{\emptyset, \dots\} = \emptyset$ will definitely be a member of set w.o. intersection But ' $\emptyset$ ' need not be the only element

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$

eg: let $S = \{1, \{1\}\}$

$P(S) = \{\emptyset, \{1\}, \{1, \{1\}\}, \{1, \{1, \{1\}\}\}$

$P(P(S)) = \{\emptyset, \{\emptyset\}, \{\{1\}\}, \dots\} \rightarrow n = \{\emptyset, \{1, \{1\}\}\}$

How "PYQ"



Q1. Let $P(S)$ denote the power set of a set S . Which of the following is always true?

~~A.~~ $P(P(S)) = P(S)$

~~B.~~ $P(S) \cap P(P(S)) = \{\emptyset\}$

~~C.~~ $P(S) \cap S = P(S)$

~~D.~~ $S \notin P(S)$ (Always false)

$S \in P(S)$ is always true

$$|A \cap B| \leq \min(|A|, |B|)$$

$$|P(S)| > |S|$$

$$\therefore |P(S) \cap S| \leq |S|$$

$$\text{and } |S| < |P(S)|$$

$\therefore P(S) \cap S$ can never be $P(S)$

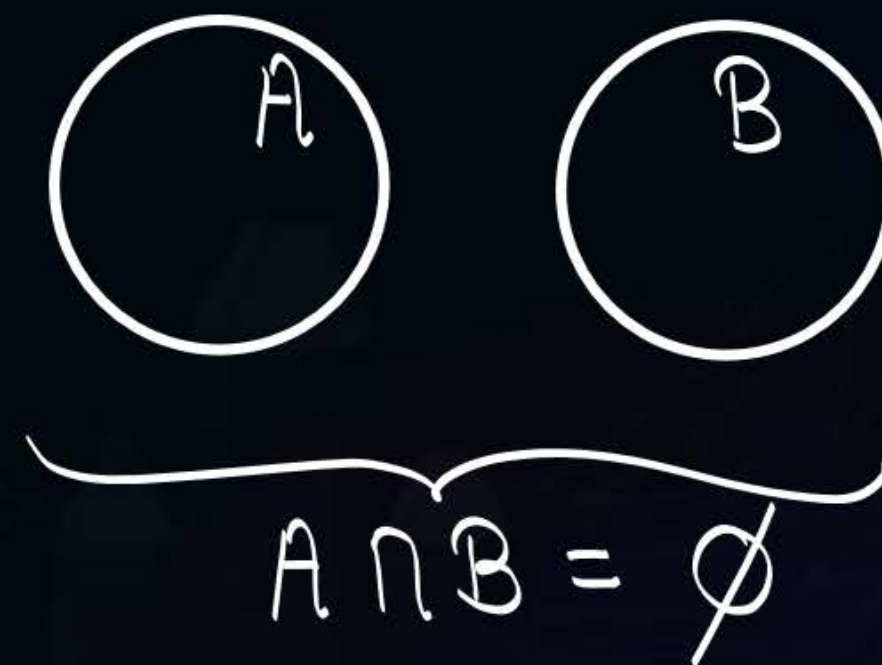


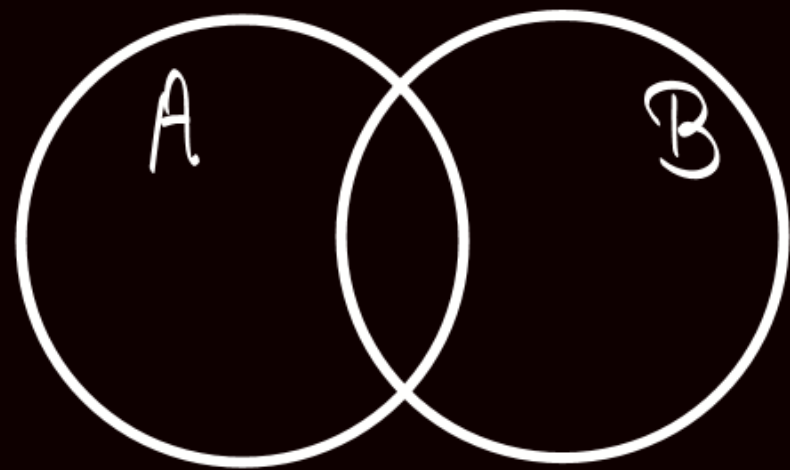
Topic : Venn Diagram

Venn diagram is used to represent the relationship among the sets pictorially.



Here $B \subseteq A$
 $A \cap B = B$
 $A \cup B = A$





$$\underbrace{\hspace{10em}}_{A \cap B \neq \emptyset}$$



Topic : Set Operations



- ✓ ☐ Complement of a set
- ✓ ☐ Union of two sets
- ✓ ☐ Intersection of two sets
- ✓ ☐ Set difference
- ✓ ☐ Symmetric difference of two sets

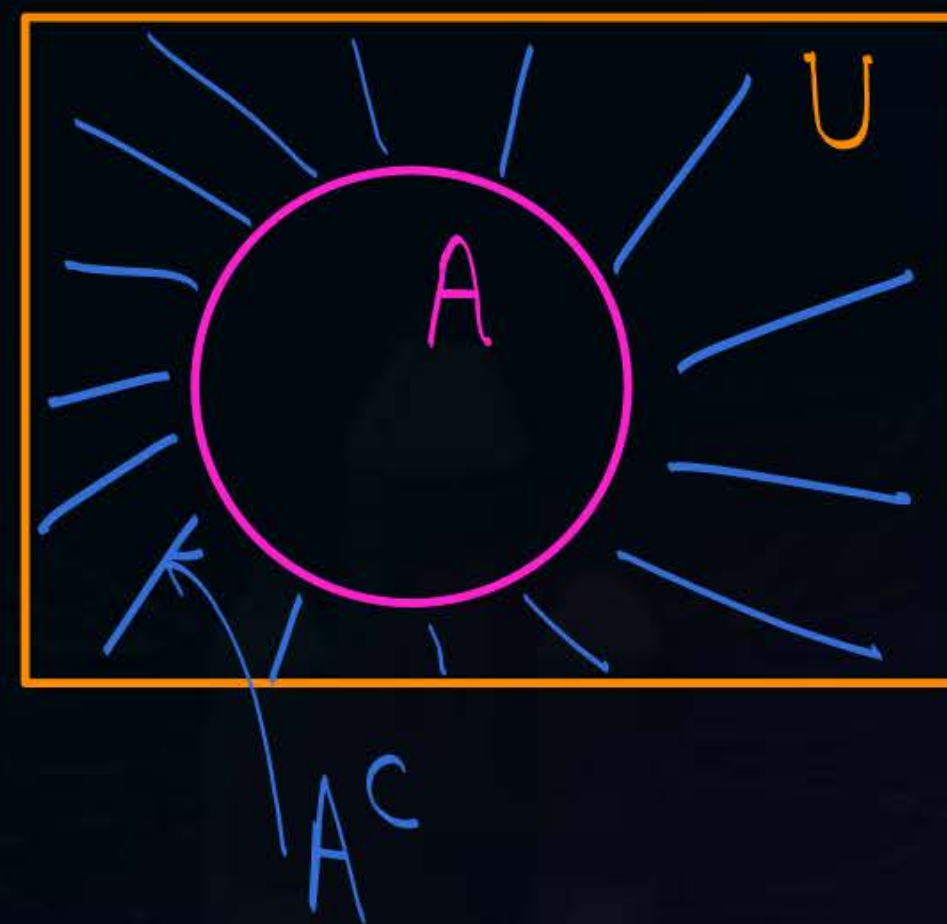


Topic : Complement of a set

Complement of a set is always defined w.r.t. universal set.

$$A^c = \{x \mid x \in U \text{ and } x \notin A\}$$

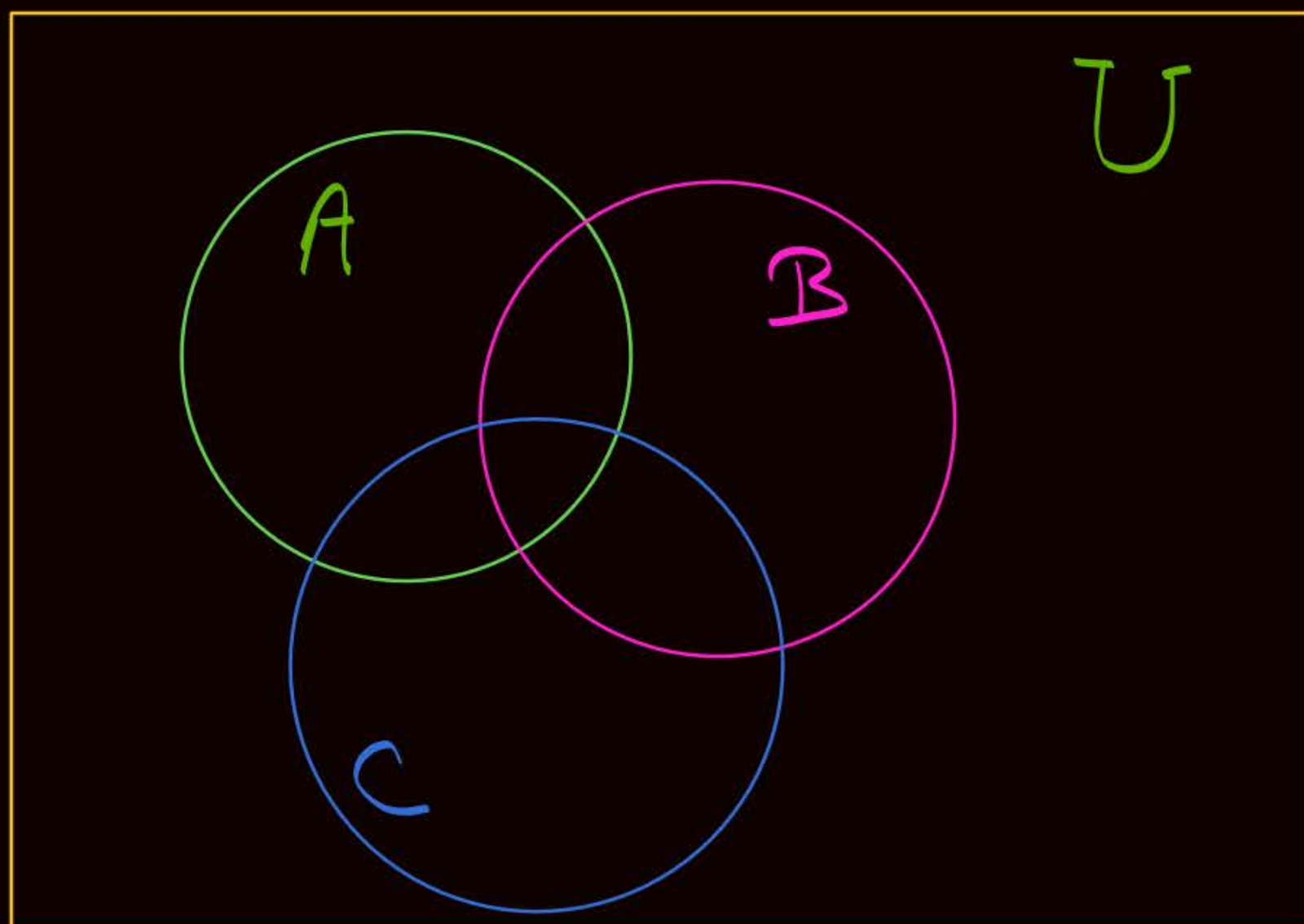
Complement of Set A



$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 4, 7, 8, 9\}$$

$$A^c = \{3, 5, 6, 10\}$$

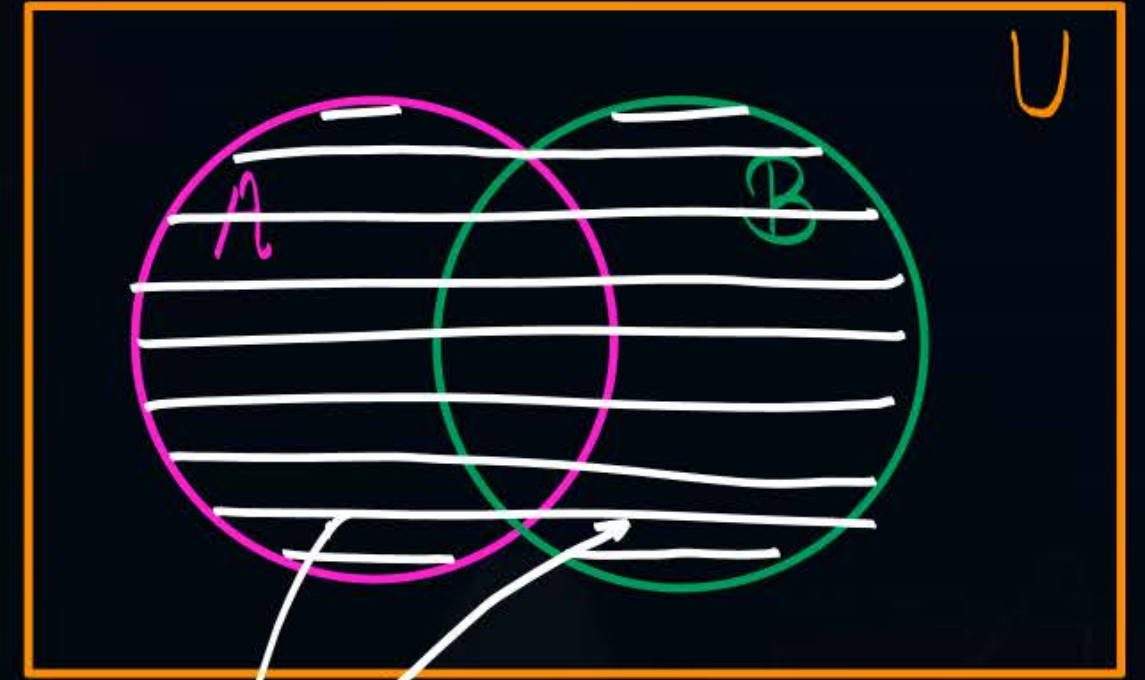




Topic : Union of two sets

Let A and B are two sets
then

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$



$A \cup B$

✓ Note: $A \cup B = B \cup A$



Topic : Intersection of two sets

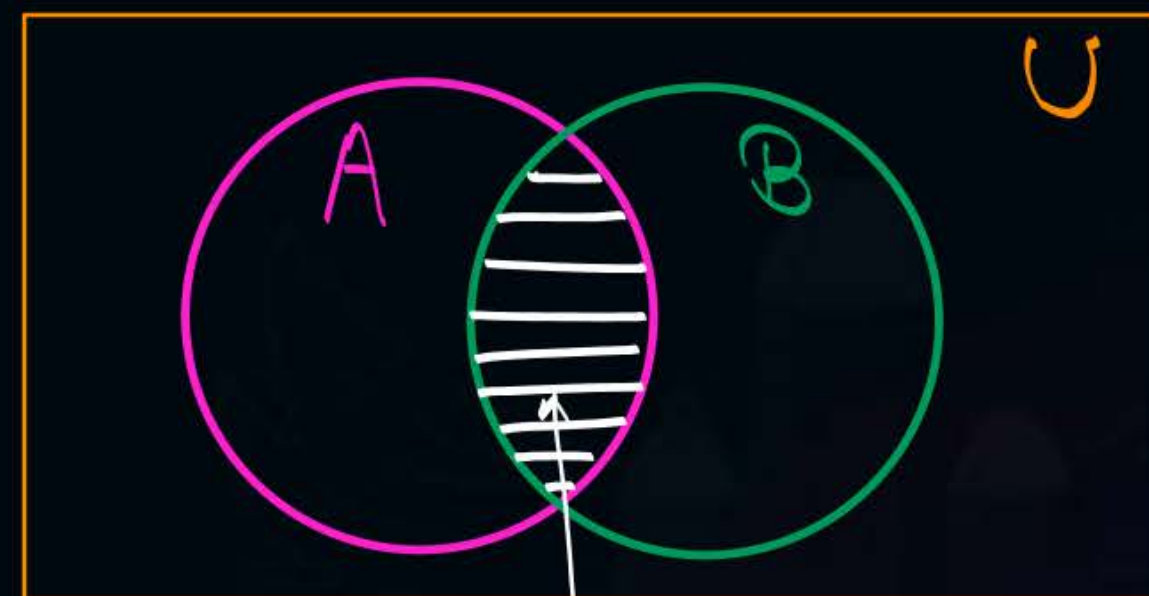
Let A & B are two sets

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$



$$A \cap B = \emptyset$$

If $A \cap B = \emptyset$, then A & B are called disjoint sets.



$$A \cap B$$

Note: $A \cap B = B \cap A$

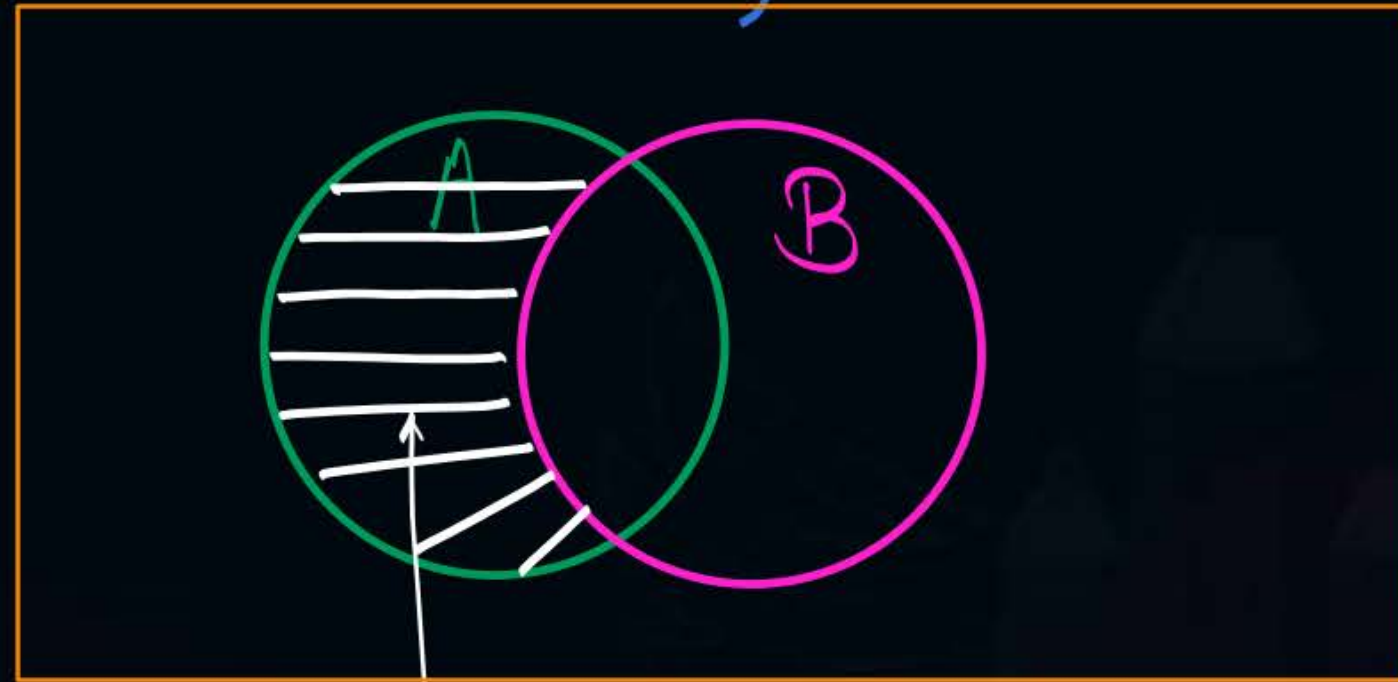


Topic : Set difference

Let A & B are two sets

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

it will result
the elements
that are present
in set A , but
not present in
set B



$$A - B = A - (A \cap B)$$

$$A - B = A \cap B^c$$

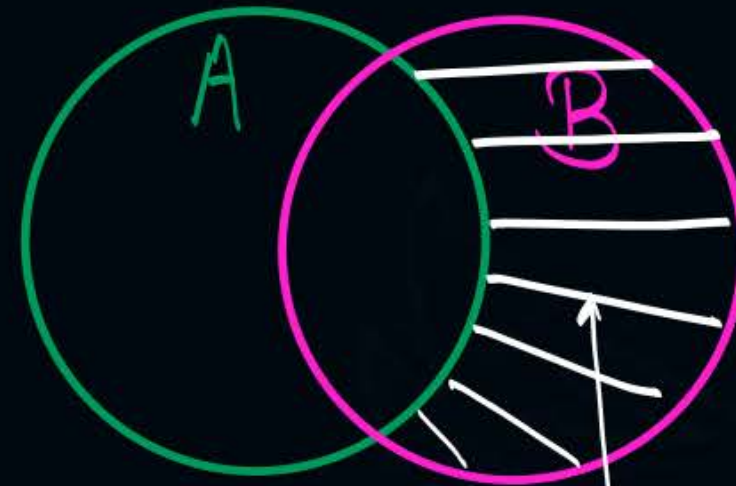


Topic : Set difference



Let A & B are two sets

$$B - A = \{x \mid x \in B \text{ and } x \notin A\}$$



$$B - A = B - (A \cap B)$$

$$B - A = B \cap A^c$$

Note: In general,

$$A - B \neq B - A$$

→ $A - B = B - A$, if and only if $A = B$
and in that case $A - B = B - A = \emptyset$

Note:

$$(A - B) \cap (B - A) = \emptyset$$

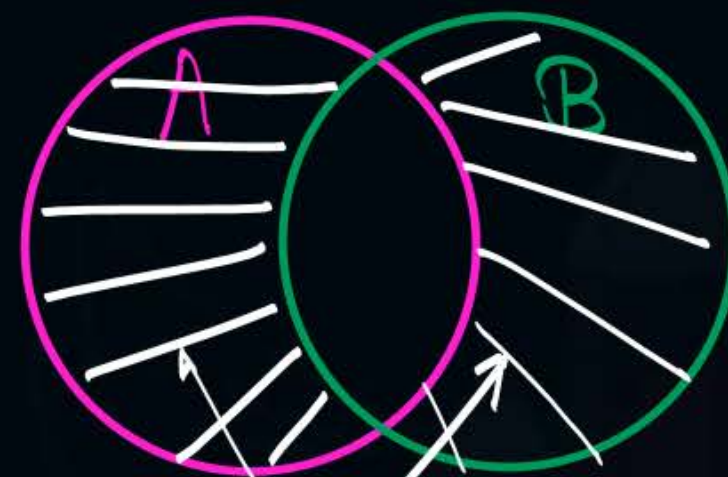
Always True



Topic : Symmetric difference of two sets

Let A & B are two sets

$$A \Delta B = \{x \mid (x \in A \text{ or } x \in B) \text{ and } x \notin A \cap B\}$$



$$\begin{aligned} A \Delta B &= (A - B) \cup (B - A) \\ &= (A \cup B) - (A \cap B) \end{aligned}$$

Note : $A \Delta B = B \Delta A$



Topic : Properties of Set Operations



w.r.t. digital logic

$$U \equiv +$$

$$\cap \equiv \cdot$$

$$U \equiv T \quad \& \quad \emptyset \equiv F$$

Universal set

✓ 1. Idempotent:

a. $A \cap A = A$

b. $A \cup A = A$

✓ 2. Identity:

a. $A \cup \emptyset = A$

b. $A \cap U = A$

$\emptyset$ is identity w.r.t. Union

Universal set is identity w.r.t. Intersection



Topic : Properties of Set Operations

3. Domination:

a. $A \cap \emptyset = \emptyset$

$\emptyset$ dominates

b. $A \cup U = U$

U dominates



Topic : Properties of Set Operations

4. Complementation:

✓ a. $A \cup A^c = U$

✓ b. $A \cap A^c = \emptyset$



5. Double Complement:

a. $(A^c)^c = A$



Topic : Properties of Set Operations

✓ 6. Commutative

a. $A \cup B = B \cup A$

b. $A \cap B = B \cap A$

✓ 7. Associative

a. $A \cup (B \cup C) = (A \cup B) \cup C$

b. $A \cap (B \cap C) = (A \cap B) \cap C$



Topic : Properties of Set Operations

8. Absorption

a. $A \cup (A \cap B) = A$

$(A \cap B) \subseteq A$ as well as $(A \cap B) \subseteq B$

b. $A \cap (A \cup B) = A$

$A \subseteq (A \cup B)$ as well as $B \subseteq (A \cup B)$

9. DeMorgan's

a. $(A \cup B)^c = A^c \cap B^c$

b. $(A \cap B)^c = A^c \cup B^c$



Topic : Properties of Set Operations

10. Distributive

a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$



Topic : Multi-set

- An unordered collection of objects/elements in which an element may appear more than once is called a multiset.

eg: Multiset $A = \{a, b, b, a, a, c, d, c\}$

Size of multiset $A = 8$

$= \{3.a, 2.b, 2.c, 1.d\}$

These numbers will represent the multiplicity of the corresponding elements

$= 3 + 2 + 2 + 1 = 8$

Let multiset $P = \{m_1 \cdot a_1, m_2 \cdot a_2, m_3 \cdot a_3, \dots, m_k \cdot a_k\}$
Where ' m_i ' is the multiplicity of element a_i .

And
let multiset $Q = \{n_1 \cdot a_1, n_2 \cdot a_2, n_3 \cdot a_3, \dots, n_k \cdot a_k\}$
Where ' n_i ' is the multiplicity of element a_i .

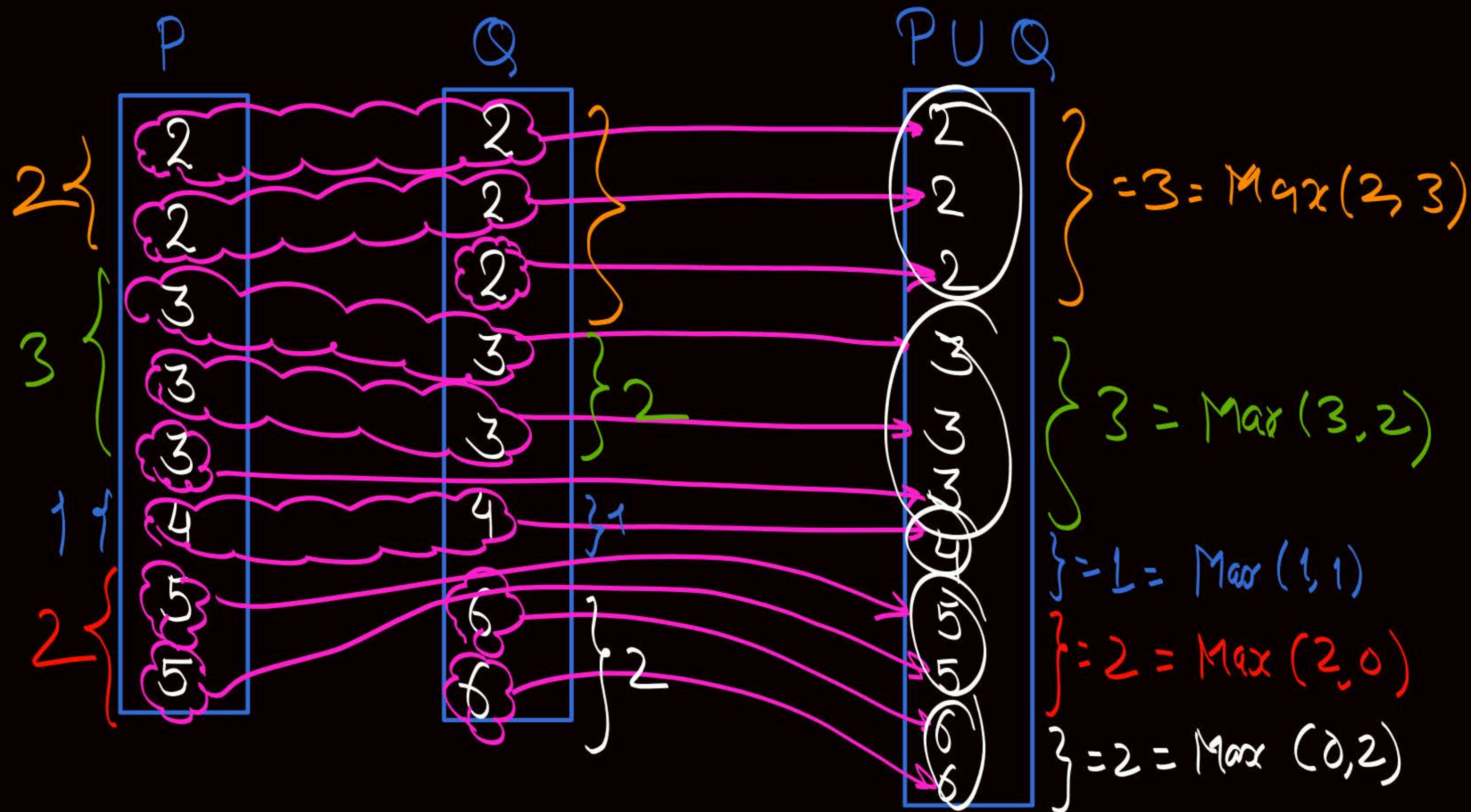
Find multiplicity of a_i in

(i) $P \cup Q$

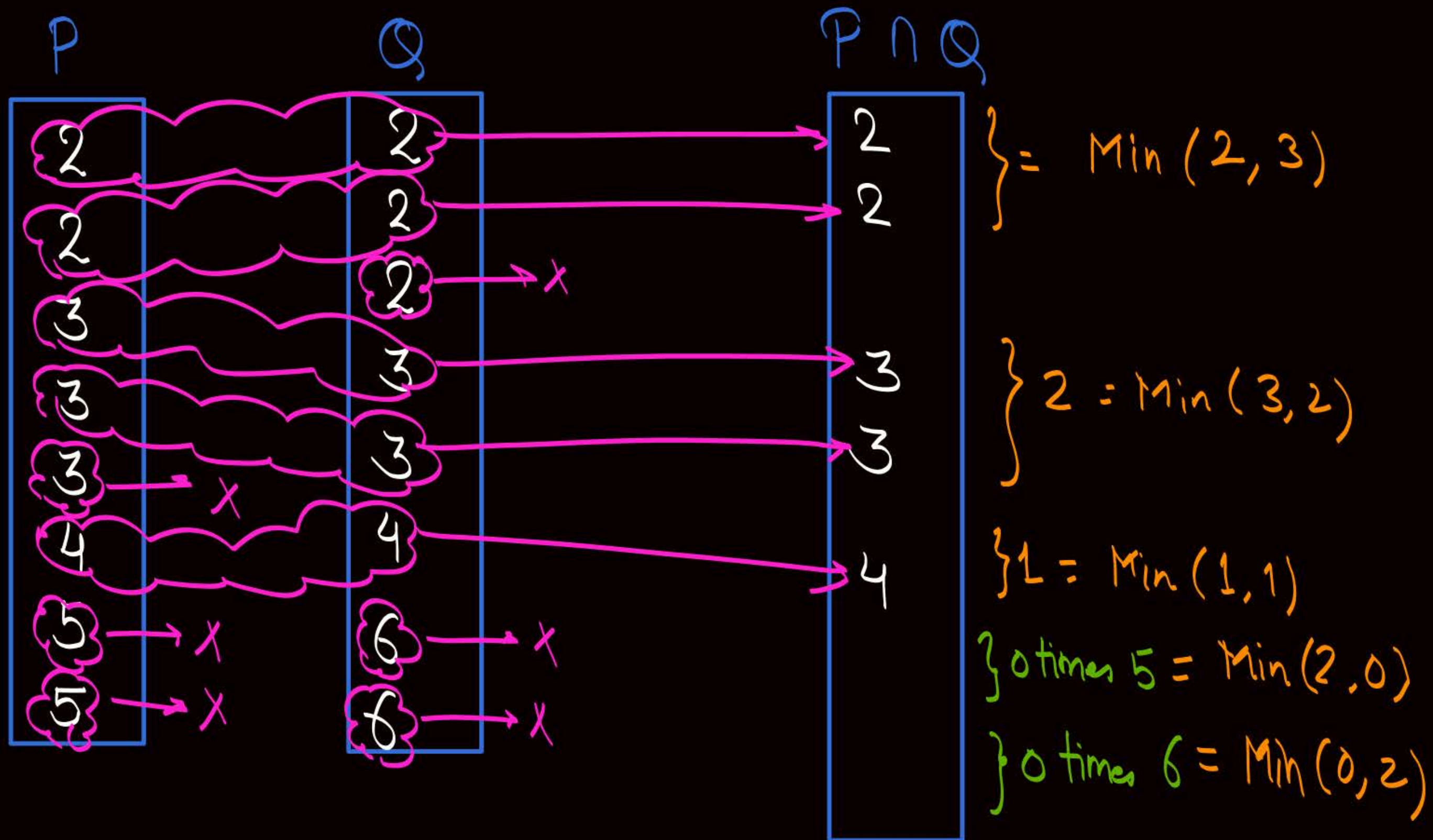
(ii) $P \cap Q$

(iii) $P - Q$

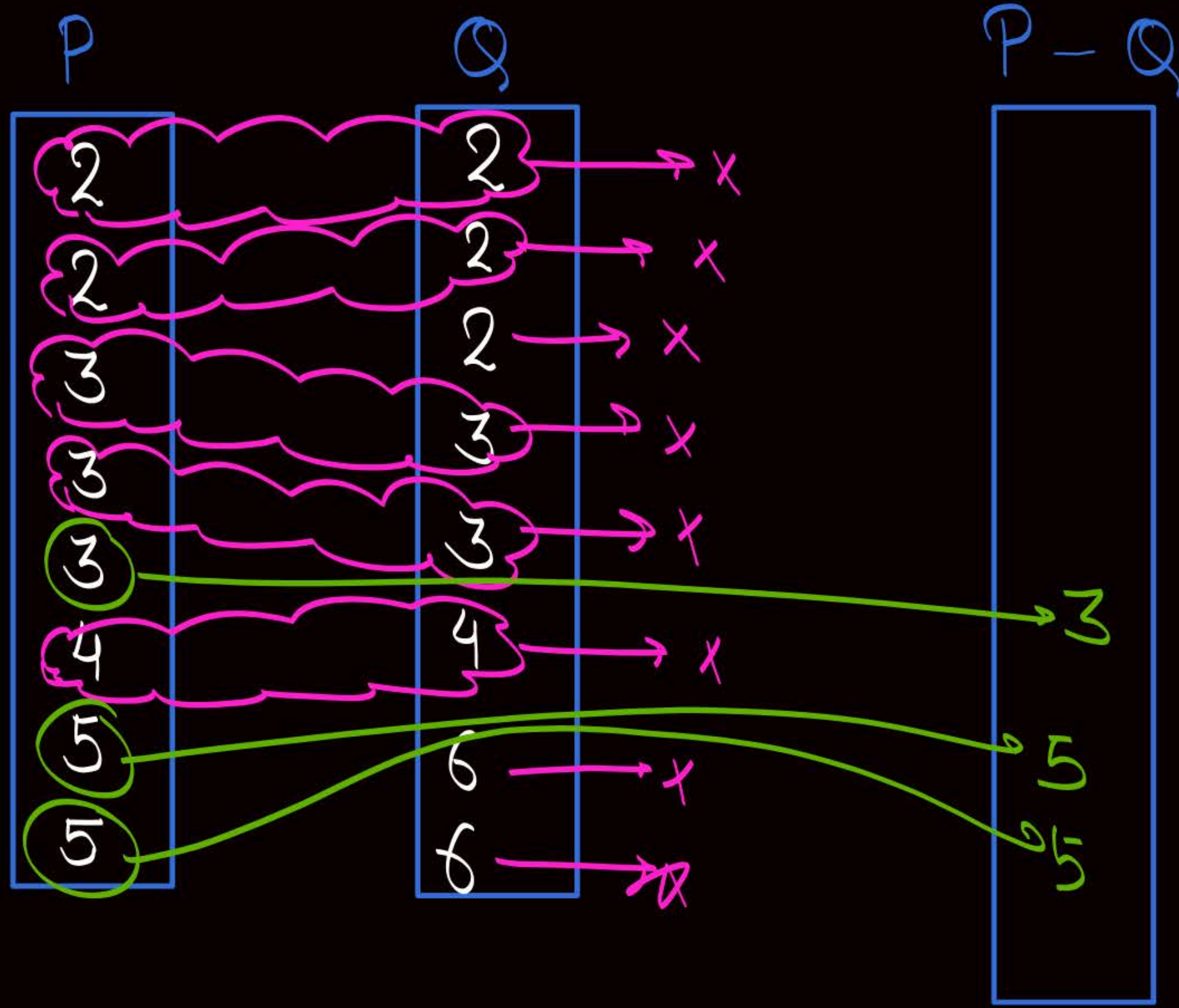
Multiset $P \cup$ Multiset Q



Multiset $P \cap$ Multiset Q



Multiset P — Multiset Q



Let multiset $P = \{m_1.a_1, m_2.a_2, m_3.a_3, \dots, m_k.a_k\}$
Where ' m_i ' is the multiplicity of element a_i .

And
let multiset $Q = \{n_1.a_1, n_2.a_2, n_3.a_3, \dots, n_k.a_k\}$
Where ' n_i ' is the multiplicity of element a_i .

Find multiplicity of a_i in

(i) $P \cup Q$ = multiplicity of a_i in $P \cup Q = \max(m_i, n_i)$

(ii) $P \cap Q$ = multiplicity of a_i in $P \cap Q = \min(m_i, n_i)$

(iii) $P - Q$ = multiplicity of a_i in $P - Q = \begin{cases} (m_i - n_i) & \text{if } m_i > n_i \\ 0 & \text{otherwise} \end{cases}$

Q: Let $A = \{1, 2, 3, 4, \dots, n\}$

How many different multisets can be created using the element of set A. { Any element of set A
Can be taken any no. of times }

$$M_1 = \{1\}$$

$$M_2 = \{1, 1\}$$

$$M_3 = \{1, 1, 1\}$$

$\vdots$

$$M_{100} = \{1, 1, 1, \dots, 1\}$$

100 times '1'

$\vdots$

$$M_\infty = \{1, 1, \dots\}$$

infinite times 1

If we do not restrict the size of multisets then we can create infinite multisets using a single element

$\therefore$ Ans = infinite

Q: Let $A = \{1, 2, 3, 4, \dots, n\}$
How many different multisets of size = 4 can be
formulated using the elements of set A, such that
at least one element of set A appears exactly
twice in each such multiset.

Based on the given statement only two types of multisets
 are possible.

$A = \{1, 2, 3, \dots, n\}$

Type-1 $\{a, a, b, c\}$

$\{a, b, a, c\}$

$${}^nC_1 \times \frac{{}^{(n-1)}C_1 \times {}^{(n-2)}C_1}{2!} = \frac{n(n-1)(n-2)}{2}$$

$$({}^nC_1) * ({}^{(n-1)}C_2) = \frac{n(n-1)(n-2)}{2}$$

$\{a, a, b, b\}$ Type-2

$\{a, b, b, a\}$

$$\frac{{}^nC_1 \times {}^{(n-1)}C_1}{2} = \frac{n(n-1)}{2}$$

$$({}^nC_2 * 1) = \frac{n(n-1)}{2}$$

a, b, c

a, a, b, c

b, b, a, c

c, c, a, b

$${}^nC_3 * 3 * 1 * 1 = \frac{n(n-1)(n-2)}{3 \cdot 2 \cdot 1}$$

$$= \frac{n(n-1)(n-2)}{2}$$

Q: Let $A = \{1, 2, 3, 4, \dots, n\}$
How many different multisets of size = 4 can be formulated using the elements of set A, such that at least one element of set A appears exactly twice in each such multiset.

* Required multiset may be of Type-1 or Type-2

∴ Total no. of Multisets of required type =

No. of multisets using Type-1 + No. of multisets using Type-2

$$= \frac{n(n-1)(n-2)}{2} + \frac{n(n-1)}{2} = \boxed{\frac{n(n-1)^2}{2}}$$

H.W. Q: Let $A = \{1, 2, 3, 4, \dots, n\}$
How many different multisets of 'size = 4' can be
formulated using the elements of set A.



2 mins Summary



✓
Topic

Venn diagram

✓
Topic

Set operations and properties of set operations

✓
Topic

Multi-set

THANK - YOU